THE JOURNAL OF PHYSICAL CHEMISTRY LETTERS
A JOURNAL OF THE AMERICAN CHEMICAL SOCIETY

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Letter

A Polarizable Cationic Dummy Metal Ion Model

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Cite This: J. Phys. Chem. Lett. 2022, 13, 5334−5340

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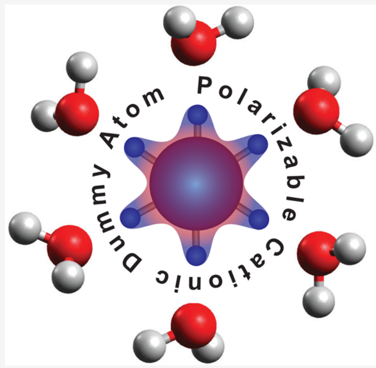
ABSTRACT: A novel locally polarizable multisite model based on the original cation dummy atom (CDA) model is described for molecular dynamics simulations of ions in condensed phases. Polarization effects are introduced by the electronegativity equalization model (EEM) method where charges on the metal ion and its dummy atoms can fluctuate to respond to the environment. This model includes explicit polarization and ion-induced interactions and can be coupled with nonpolarizable or polarizable water models, making it more transferable to simpler force fields. This approach allows us to enhance the original fixed charge CDA model where the charge distribution cannot adapt to the local solvent structure. To illustrate the new CDA^{pol} model, we examined properties of the Zn^{2+}, Al^{3+}, and Zr^{4+} ions in aqueous solution. The polarizable model and Lennard-Jones parameters were refined for octahedrally coordinated Zn^{2+}, Al^{3+}, and Zr^{4+} CDAs to reproduce thermodynamic and geometrical properties. Using this locally polarizable model, we were able to obtain the experimental hydration free energy, ion−oxygen distance, and coordination number coupled with the standard 12−6 Lennard-Jones model. This model can be used in myriad additional applications where local polarization and charge transfer is important.



Metal ions play significant roles in biochemistry and materials science. Around one-third of the proteins in the protein databank contain metal ions.[1] Metal ions form complexes with surrounding amino acids and serve significant functional roles in biological systems,[2] including structure,[3] electron transfer, and catalysis.[4] Modeling of systems including transition metals represents one of the more challenging tasks using standard classical simulations. The charge on transition metals is not constant and is affected by factors like the environment, nature of coordinated ligands, and oxidation state of the metal. There is no general solution to date for the representation of transition metals in molecular force fields; as such, this is still a matter of ongoing research and development. The electronic cloud is usually distributed nonsymmetrically around the metal ion and can further change and redistribute in response to changes in the surrounding environment (Figure 1). Transition metals have d and f orbitals as their outermost orbitals, which permit complicated chemical bonding patterns.

Classical molecular dynamics (MD) simulations are mostly based on fixed atomic charge models that do not allow the electrostatic properties to change as a function of geometry nor respond to electrostatic potentials generated by nearby atoms. The simple idea of electrostatic interactions using fixed-point charges is not entirely successful in many circumstances because electrons are spread out in space and atomic charges vary continuously. For instance, properties of metal ions in aqueous solutions are not well-modeled by simple Lennard-Jones potentials which make the approximation that the ion has a formal charge.[5] Lennard-Jones potentials do not account for ion-induced dipole interactions, while fixed charge models do not account for polarization effects. These limitations have

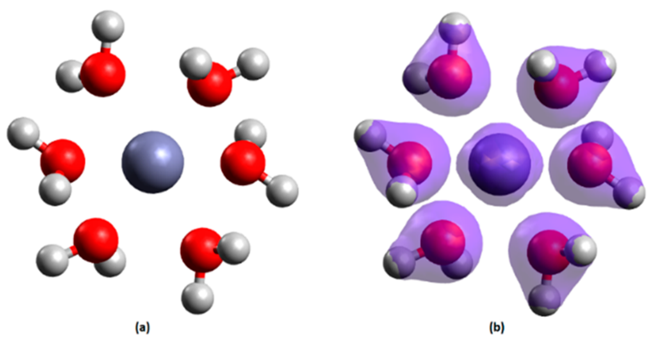
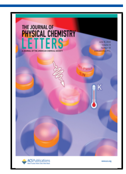


Figure 1. Polarizable models achieve higher fidelity than classical atom-centered fixed charge models through more accurate electrostatic interactions which leverage dynamically updated charge distributions. Depicted is a middle ion coordinating with six water molecules in (a) classical description and (b) with electron density distribution.

Received: April 28, 2022
Accepted: June 3, 2022
Published: June 8, 2022



led to the creation of a range of polarizable force fields.[6] In addition to the fluctuating charge model, two main categories of polarizable models are the Drude models[7,8] and the induced dipole models.[9] In the Drude model, each polarizable atom is represented by a pair of point charges, in which a Drude particle is attached to the corresponding atomic site with a spring parametrization mimicking nuclear−electron motion. Also, the induced dipole model is constructed based on multipole expansion of a charge distribution, which is often truncated at the dipole term in the induced dipole model. The AMOEBA[10] polarizable force field is a well-known model in this category that represents using multipole moments that include not only monopoles but also higher-order terms such as dipoles and quadrupoles.

Polarizable models with proper tuning can accurately predict charge distributions in molecular simulations; as such, incorporation of polarization effects to traditional force fields is an ongoing effort.[6,8,11−14] Polarizable force fields allow the electrostatic variables to change as a function of geometry or in response to the electrostatic potentials generated by nearby atoms and update the charges accordingly during a simulation. Applications of polarizable force fields to various problems such as protein−ligand binding,[11] pK_a prediction,[12] and ion solvation[13] have been reported. Despite the benefits of polarizable models, they are oftentimes more computationally demanding because of their complexity. That being said, with continuing improvements in the speed of computational hardware, more accurate and consequently more computationally expensive simulations can be afforded. While quantum chemistry methods can be used to predict and update charges in more accurately and reliably, quantum mechanics (QM) methods are still limited to much smaller systems simulated over shorter time scales compared to polarizable models. The computational cost of the calculations within the QM region also remains the rate-limiting factor in hybrid quantum mechanics/molecular mechanics (QM/MM) simulations. In this regard, polarizable force fields offer significant opportunities at a reasonable increase in the computational cost of a molecular simulation.

In this Letter, we utilize the recently developed hybrid ReaxFF/AMBER MD tool,[15] which introduces hybrid polarizable/nonpolarizable force field simulations via the QM/MM interface within the AMBER MD package and is freely available in AMBERTools 21 and beyond.[16] The ReaxFF/AMBER MD tool has been implemented by developing an efficient interface to the open source PuReMD software[17] and uses the electronegativity equalization model (EEM) within PuReMD for charge equilibration. While no ReaxFF simulations were performed in this work, the ReaxFF/AMBER tool allows us to utilize the polarization capability of EEM within the standard 12−6 Lennard-Jones model. In EEM, one electronegativity and one hardness parameter are assigned to each element, and these parameters are optimized against QM-charge distributions. The electrostatic energy is expressed as a function of the electronegativity and hardness parameters:

$$E_{\text{elec}} = \sum_i \left(\chi_i q_i + \frac{1}{2}\eta_i q_i^{\,2}\right) + E_{\text{Coulomb}} \tag{1}$$

In eq 1, the term in the summation over all atoms i gives the self-energy contributions where χ represents the Mulliken electronegativity of an atom and η represents twice the Parr−Pearson hardness. In the polarizable potential used in this study, Coulomb interaction with electrostatic shielding is adopted to prevent excessively high repulsion between nearby atoms as shown in eq 2, where C is the Coulomb constant in vacuum, q_i and q_j denote the atomic charges, r_{ij} is the distance between atoms i and j, and γ_{ij} is the electrostatic shielding between the elements corresponding to atoms i and j.

$$E_{\text{Coulomb}} = C\cdot\frac{q_i q_j}{\sqrt[3]{r_{ij}^{\,3} + \left(\frac{1}{\gamma_{ij}}\right)^3}} \tag{2}$$

The EEM model is based on atom-centered point charges, whereas a more realistic charge distribution should be smooth and anisotropic. The details of EEM method can be found in the Supporting Information. To capture the anisotropic features, induced dipole models and/or off-center sites can be added to obtain a hybrid polarizable model. The cationic dummy atom (CDA) model of Åqvist and Warshel[18] is an attractive way to add multisite charge distribution to a metal ion atom center. An octahedral geometry is the most common topological scheme for CDA simulations, but other topological schemes like tetrahedral, tetrahedral-like with three dummy atoms and an empty site, or pentagonal bipyramidal have also been used.[19,20] A tetrahedral CDA model with zinc as the central metal ion was used to evaluate the binding free energy of zinc to proteins, and the free energy of binding contributed by the conformational fluctuations of the zinc-binding site was enhanced using this model compared to the standard, spherical ion model.[21] Pentagonal bipyramidal CDA models were used to improve the solvation, binding thermodynamics, and coordination geometries of Ca^{2+} and Mg^{2+} ions.[20] Octahedral CDA models have been used for enhancing the modeling of systems including Mg^{2+},[22] Cu^{2+},[23] Mn^{2+},[24] Ni^{2+},[25] and more.[26] While these models use the standard 12−6 Lennard-Jones (LJ) potential and neglect polarization and ion-induced dipole interactions, attempts to develop a new type of LJ potential (12−6−4) by adding an additional $\frac{1}{r^4}$ term[27] to describe the ion-induced dipole interactions for ions represented by standard single atom[28] and ions represented by CDA[29] have been reported.

In this work, we describe a novel CDA model, called CDA$^{\text{pol}}$, where polarization and ion-induced electrostatic interactions are explicitly incorporated through the dynamic charges calculated using EEM. The CDA$^{\text{pol}}$ model is similar to the Drude model[8,30] in spirit in which part of the mass and charge of the ion is fractioned into multiple sites anchored to the ion center. The CDA model delocalizes a part of the total charge on dummy atoms placed around the central ion to mimic covalent bonds. While reproducing properties of highly charged ions using fixed charge models has been challenging, we demonstrate that the novel CDA$^{\text{pol}}$ model addresses many of these issues and outperforms the standard CDA model. To illustrate CDA$^{\text{pol}}$, we report results for exemplar metal ions (Zn^{2+}, Al^{3+}, and Zr^{4+}) using both the standard CDA and the CDA$^{\text{pol}}$ models. Six fractional positive charges were added to dummy atoms which are connected to the middle ion in an octahedral geometry at a constant distance (specifically, 0.9 Å); the nonbonded CDA interactions include one LJ center on the central atom and seven (one middle ion and six dummy atoms) electrostatic centers on all CDA atoms. The charges within the CDA$^{\text{pol}}$ model were updated at every step with a

https://doi.org/10.1021/acs.jpclett.2c01279
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constraint on the total charge of +2, +3, and +4. In CDA^{pol} models, separate electronegativity and hardness parameters were assigned to the middle ion and dummy atoms; these parameters were then optimized to allow the charges to fluctuate within the partial and total charge constraints specified in the CDA^{pol} model. For nonpolarizable CDA models, just one AMBER force field was required for performing solvation simulations, while for polarizable CDA^{pol} models in addition to AMBER force field, a polarizable force field was required. If the polarizable model is used, in addition to the AMBER potential, a separate potential is required for modeling the polarizable region. Consequently, two separate parameter optimization steps are needed. First, the polarizable potential parameters were tuned to reproduce the target charge distributions on CDA^{pol} and DFT training data obtained from ion hydration. These force field trainings were performed by optimizing the middle ion and dummy atoms' EEM electronegativity, hardness, and electrostatic interactions shielding parameters in addition to ion−oxygen and ion−hydrogen interaction parameters. Second, 12−6 LJ interaction parameters were tuned to reproduce the target hydration free energies (HFE) and ion−oxygen distance (IOD) values (Figure 2).

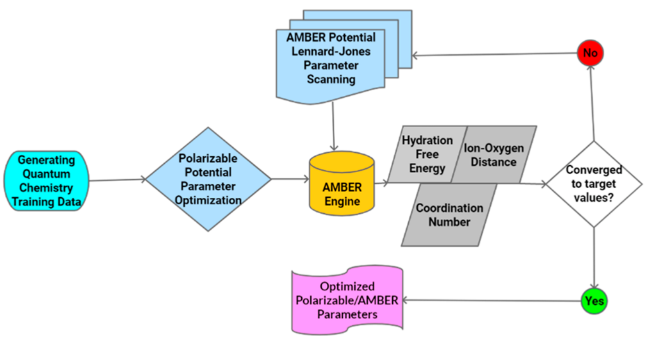

Figure 2. Flowchart for the CDA^{pol} potential training and modeling.

During LJ parameter scans, the parameter ε was varied from 1 to 5 kcal/mol in increments of 0.2, while the $R_{min}/2$ parameter was scanned with a 0.1 Å interval. Further details regarding the parameter optimization procedures used in this work can be found in the Supporting Information.

Using this locally polarizable model in combination with TI calculations, we can examine the effect of the CDA^{pol} model charge updates in response to the environmental changes. To evaluate the fluctuations in CDA charges, the charges on the middle ion and dummy atoms were collected during a 50 ps equilibration of the solvated CDA ion. The charge fluctuations on the middle ion and dummy atoms are shown in Table 1 and in Figure S.3 in the Supporting Information. These data show that although the charges on the middle ion and dummy atoms fluctuate near the target value, more deviation on the Al^{3+} middle ion is observed toward negative values. This tendency can be attributed to the combination of total charge and target IOD included in the training set for the polarizable force field. For Al^{3+} CDA^{pol}, because the target IOD (1.88 Å) is smaller than two other ions (2.1 Å for Zn^{2+} CDA^{pol} and 2.2 Å for Zr^{4+} CDA^{pol}), the system tends to shift to more positive charges on dummy atoms and consequently to a negative charge on the middle ion. The dipole moment fluctuations for CDA^{pol} during 50 ps *NPT* equilibration runs are shown in Figure 3. The details of the calculation of dipole moments can be found in section S.3 of the Supporting Information. The averages of 1000 snapshot points are 0.32D, 0.22D, and 0.53D for Zn^{2+}, Al^{3+}, and Zr^{4+} respectively, which is not an unexpected value for the CDA^{pol} models.

Three-point Gaussian integration of free energy difference of two states yielded the HFE shown in Figure 4. These contour graphs include the final HFE values in the LJ scanning space using the nonpolarizable CDA and polarizable CDA^{pol} models. When using the CDA model, HFE values vary in a wider range compared to when using CDA^{pol}. While the presence of polarization reduces the magnitude of the range of the HFE, the CDA^{pol} model shows more local structures relative to the CDA model. IOD values were calculated from radial distribution functions (RDFs) obtained with a grid resolution of 0.01 Å based on the average volume of the trajectories in the range of 0−10.0 Å, which is the same protocol previously used for calculating the IOD values of highly charged metal ions.[27] The resulting IOD values from both nonpolarizable and polarizable models are shown in Figure 5. The coordination number (CN) values were determined by integrating the ion−oxygen RDF from the origin to its first minimum in the RDF plots.

The experimental target HFE, IOD, and CN for Zn^{2+} are shown in Table 2. HFE values represent a thermodynamic property, while IOD and CN represent structural properties. Using the single-atom nonpolarizable model and the 12−6 LJ nonbonded model only (denoted as AMBER), the target HFE and IOD values were not simultaneously reproduceable. With the best calculated HFE, structural IOD and CN parameters were far from target values. As shown in Figure 6, using the CDA model can result in improvements in estimating the structural parameters. However, IOD and CN values are still away from the target values with the standard CDA model. Shifting to the CDA^{pol} model resulted in more accurate IODs

Table 1. Averages and Standard Deviations (SD) for CDA^{pol} Atomic Charges during 50 ps Equilibration Runs (1K Data Points)

Zn^{2+}				Al^{3+}				Zr^{4+}			
atom	target charge	average charge	SD	atom	target charge	average charge	SD	atom	target charge	average charge	SD
middle ion	0	0.004	0.004	middle ion	0	−0.449	0.012	middle ion	0	−0.113	0.001
D1	0.333	0.335	0.025	D1	0.5	0.574	0.017	D1	0.666	0.687	0.0507
D2	0.333	0.331	0.026	D2	0.5	0.576	0.017	D2	0.666	0.686	0.045
D3	0.333	0.331	0.027	D3	0.5	0.575	0.017	D3	0.666	0.688	0.048
D4	0.333	0.334	0.026	D4	0.5	0.575	0.017	D4	0.666	0.684	0.048
D5	0.333	0.331	0.027	D5	0.5	0.576	0.017	D5	0.666	0.685	0.046
D6	0.333	0.335	0.026	D6	0.5	0.574	0.017	D6	0.666	0.683	0.046

https://doi.org/10.1021/acs.jpclett.2c01279
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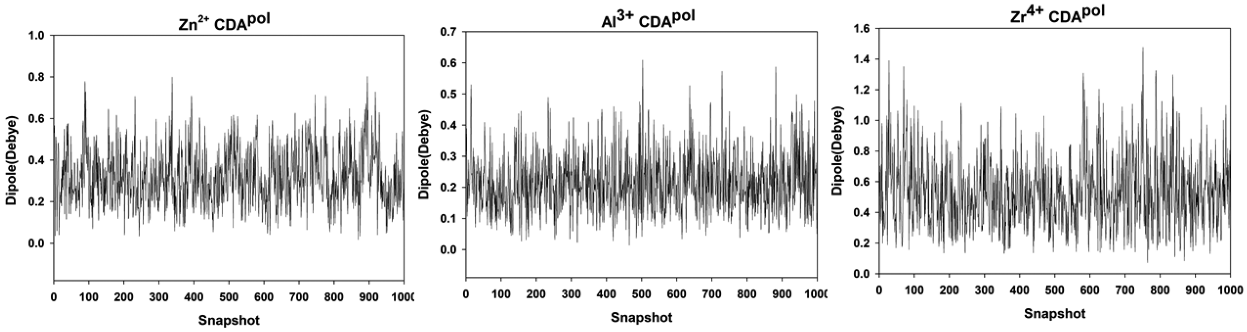

Figure 3. Instantaneous dipole moments of the CDA^{pol} molecule in solution during the simulation.

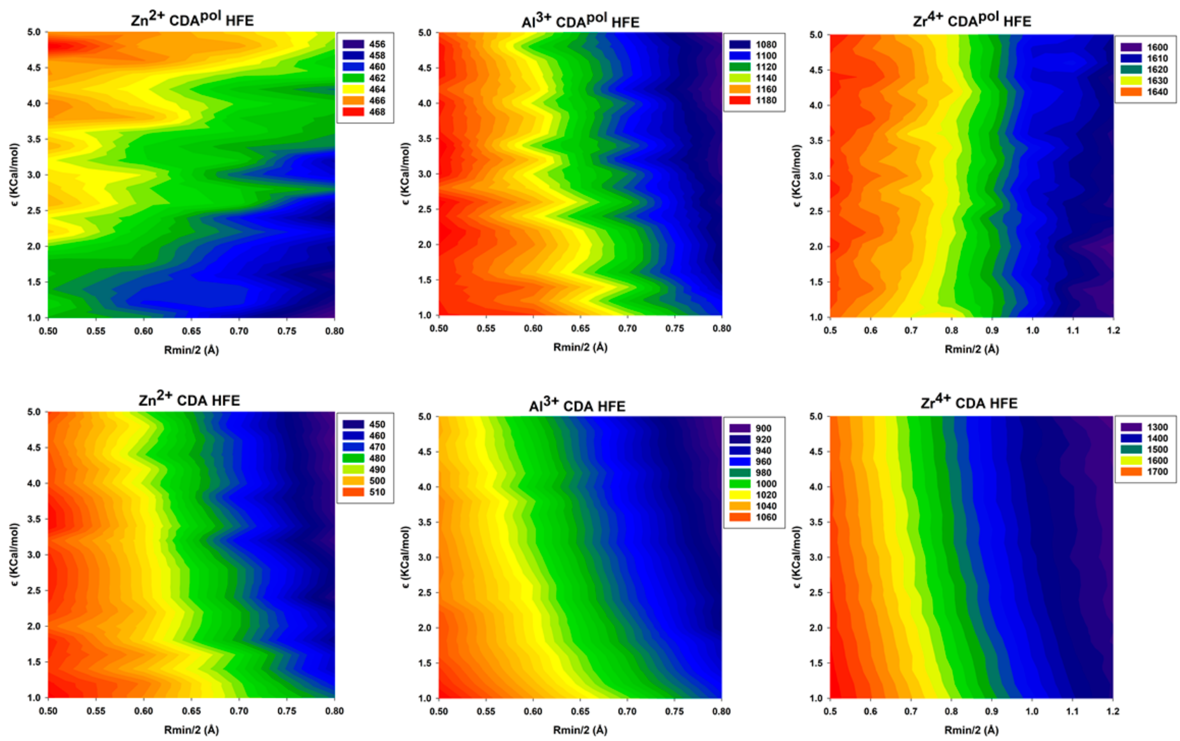

Figure 4. HFE values obtained from scanning LJ parameters. Absolute values are reported in the legends.

for all ions. The only drawback was a slight increase in the CN for Zn^{2+} CDA^{pol}, demonstrating the greater flexibility of CDA^{pol}. This shows that with the local polarizability in CDA^{pol}, while the total charge of the CDA^{pol} molecules is constrained to a constant value, the atomic charges respond to their environment yielding an instantaneous dipole that allows the thermodynamic and geometrical properties to be well-reproduced using physically meaningful LJ parameters.[31]

To obtain an estimate of the execution time of the simulations using nonpolarizable or polarizable models, the average of the total simulation time of 500 *NPT* equilibration simulations using the Simulated Annealing with NMR-Derived Energy Restraints (SANDER) program within AMBER and the SANDER ReaxFF/AMBER of the solvated CDA^{pol} were calculated. Each simulation was performed for 50 ps using a single Intel Xeon E5-2680v4 core. These benchmark calculations showed a 35% increase in computational cost when using the CDA^{pol} model. Thus, a modest increase in computational cost is sufficient to reap the benefits of a polarizable CDA model for more accurate calculations of thermodynamic and geometrical properties.

In summary, a novel cation dummy atom model with polarization effects, called CDA^{pol}, was described. CDA^{pol} includes polarization effects induced by a solvent surrounding highly charged ions as well as ion-induced dipole interactions which correspond to nonsymmetric charge distributions in the CDA^{pol} dummy atoms. In this model, a dynamic charge equilibration scheme was adopted, and extra relevant physics is added by implementing explicit ion-induced dipole interaction terms compared to the previous polarizable models. Also, this model does not necessarily require polarizable water models, making it more transferrable to simpler force fields and reducing the overall computational cost. In a proof-of-concept demonstration with the CDA^{pol} model, experimental thermodynamic and geometrical observables for hydration of Zn^{2+}, Al^{3+}, and Zr^{4+} ions were reproduced. The parametrizations were performed through a scan of the 12−6 LJ potential and electrostatic parameters of the polarizable force field. Our CDA^{pol} model reduces the HFE range, and this reduction can be attributed to the inclusion of polarization and induced dipole effects in this model. It was demonstrated that this locally polarizable model reproduces the experimental HFE,

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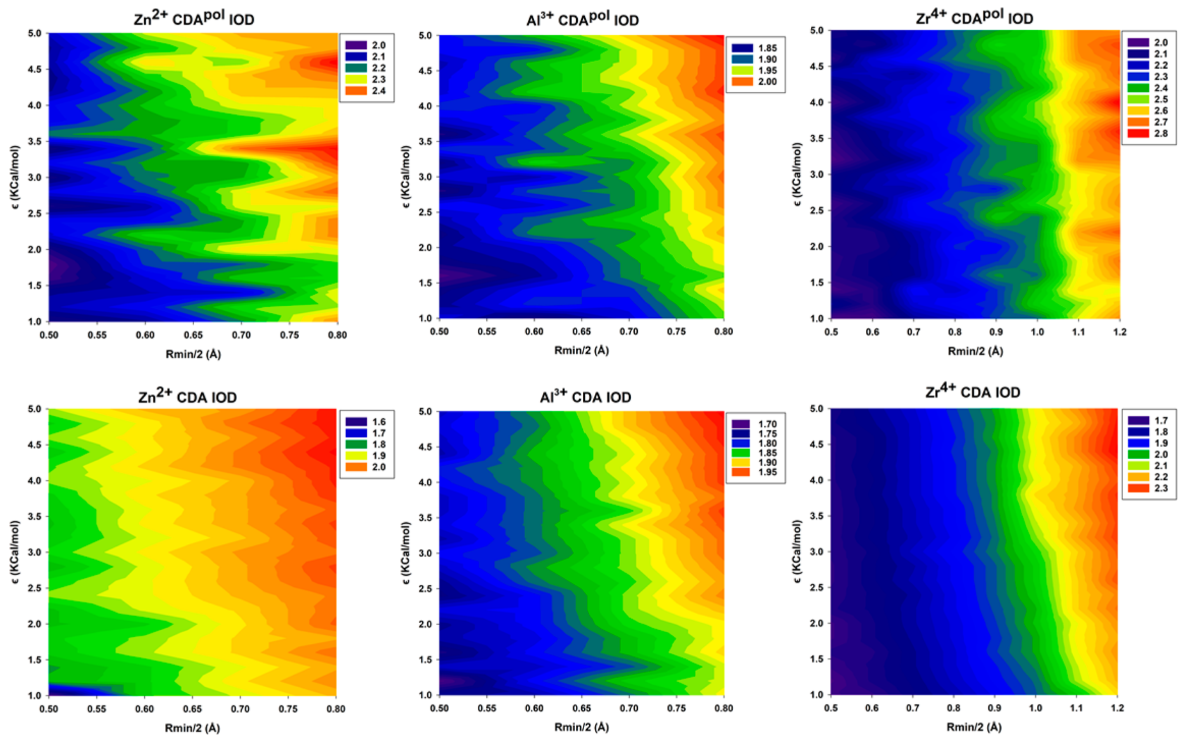

Figure 5. IOD values obtained from scanning LJ parameters.

Table 2. Target Experimental and Calculated HFE, IOD, and CN Values for Zn^{2+}, Al^{3+}, and Zr^{4+}

	Zn^{2+}				
	LJ parameters		results		
method	$R_{min}/2$	ε	HFE	IOD	CN
experiment[32]			−467.3	2.09	6.0
AMBER[a]	1.18	0.0008	−467.0	1.68	4.1
CDA	0.7	5.0	−466.2	1.96	6.0
CDA^{pol}	0.5	1.2	−462.7	2.11	6.5
	Al^{3+}				
	LJ parameters		results		
method	$R_{min}/2$	ε	HFE	IOD	CN
experiment[32]			−1081.5	1.88	6.0
AMBER[a]	0.97	6×10^{-6}	−1083.4	1.19	2.0
CDA	0.5	1.0	−1076.8	1.75	6.0
CDA^{pol}	0.8	1.6	−1083.3	1.93	6.1
	Zr^{4+}				
	LJ parameters		results		
method	$R_{min}/2$	ε	HFE	IOD	CN
experiment[32]			−1622.8	2.19	8.0
AMBER[a]	1.15	0.0005	−1622.6	1.53	4.0
CDA	0.7	1.6	−1622.0	1.77	6.0
CDA^{pol}	0.8	1.4	−1626.2	2.20	8.3

[a]Li, P. Advances in Metal Ion Modeling. Ph.D. Dissertation; Michigan State University, 2016.

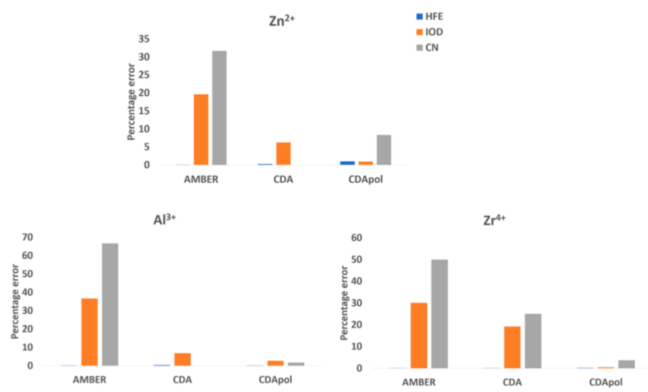

Figure 6. Percentage error of HFE, IOD, and CN calculations using classical AMBER, CDA, and CDA^{pol} with respect to target experimental values.

IOD, and CN values simultaneously with a standard 12−6 LJ potential, while this was not observed using the classic CDA model under similar modeling conditions. Using the CDA^{pol} model for the simulation of ion hydration is advantageous over fixed charge CDA and conventional spherical models because if not only allows a greater flexibility in parametrization but also is able to better reproduce the thermodynamic and geometrical observables simultaneously. The ability to capture local environmental events in large systems using this polarizable model opens a range of problems that can be tackled using CDA^{pol} to address both chemical and biological processes.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpclett.2c01279.

Polarizable and nonpolarizable potential parameter optimizations, CDA^{pol} charge fluctuations, dipole moment calculation, and electronegativity equalization method (PDF)

Transparent Peer Review report available (PDF)

https://doi.org/10.1021/acs.jpclett.2c01279
J. Phys. Chem. Lett. 2022, 13, 5334−5340

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported in part by an NIH grant (Award Number GM130641), an NSF CDS&E grant (Award Number 1807622), and an NSF OAC grant (1835144). This work used computational resources provided by the Institute for Cyber-Enabled Research at Michigan State University.

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https://doi.org/10.1021/acs.jpclett.2c01279
J. Phys. Chem. Lett. 2022, 13, 5334−5340